INCH-POUND

MIL-DTL-83413/12A
w/ AMENDMENT 1
21 December 2018
SUPERSEDING
MIL-DTL-83413/12A
09 October 2015

DETAIL SPECIFICATION SHEET

# CONNECTORS AND ASSEMBLIES, ELECTRICAL, AIRCRAFT GROUNDING: GROUNDING CLAMP CONNECTOR FOR TYPES I AND III GROUNDING ASSEMBLIES, STAINLESS STEEL CLIP, ELECTRICAL

This specification is approved for used by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-83413.

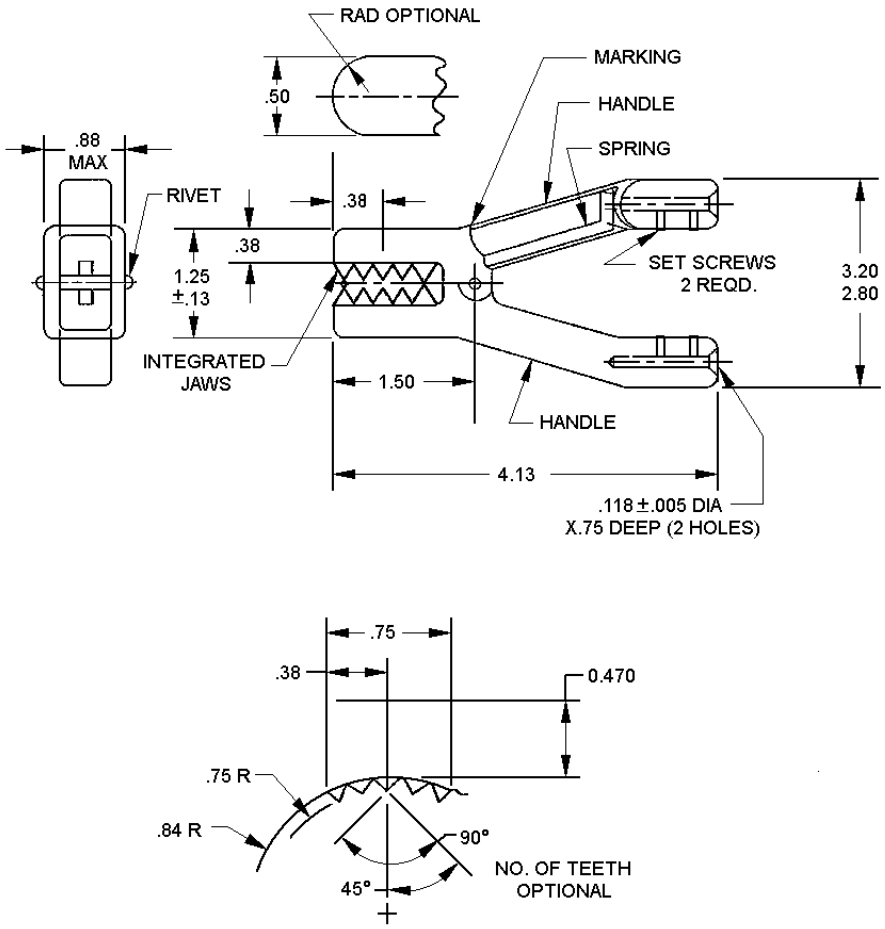



FIGURE 1. Grounding clamp, for types I and III grounding assemblies.

AMSC N/A

FSC 5999

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| Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|
| .003 | 0.08 | .113 | 2.87 | .84 | 21.3 |
| .004 | 0.10 | .125 | 3.18 | .88 | 22.4 |
| .010 | 0.25 | .130 | 3.30 | 1.25 | 31.8 |
| .031 | 0.79 | .203 | 5.16 | 1.50 | 38.1 |
| .062 | 1.57 | .205 | 5.21 | 2.80 | 71.1 |
| .078 | 1.98 | .380 | 9.65 | 3.20 | 81.3 |
| .080 | 2.03 | .500 | 12.70 | 4.13 | 104.9 |
| .090 | 2.29 | .750 | 19.1 | | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for information only.
3. Surface toughness: If cast, smooth cast finish 125 or 250 all over, except machine surface 63.
4. Remove all burrs and sharp edges .010 (0.25 mm) to .020 (0.51 mm) radius.
5. Unless otherwise specified, tolerance is ±.03 (0.8 mm) on decimals and ± 1° on angles.

FIGURE 1. Grounding clamp, for types I and III grounding assemblies - Continued.

REQUIREMENTS:

Material:

Handle: The handle material shall be stainless steel in accordance with SAE-AMS-5355, or SAE-AMS-5343, grade 17-4PH, with optional passivation.

Spring: The spring material shall be .125 diameter music wire in accordance with ASTM-A228/A228M, or equal. Heat treatment shall be as required to meet requirements. The spring shall be stress relieved, zinc/clear plated in accordance with ASTM-B633, or Nickel plated in accordance with ASTM-B733.

Set screw: The set screw shall meet the requirements of AIA/NASM51021, and shall be passivated in accordance with SAE-AMS 2700.

Salt spray: 500 hours in accordance with test procedure EIA/ECA-364-26 test condition C.

Clamp connector resistance: Shall not exceed 500 milliohms when tested in accordance with MIL-DTL-83413.

Part or Identifying Number (PIN): M83413/12-1.

QUALIFICATION: Qualification is not required. First article inspection shall consist of a visual and mechanical inspection in accordance with MIL-DTL-83413 and the requirements stated in this specification sheet.

A QPL manufacturer that periodically performs group A and B inspections on Grounding clamps in accordance with MIL-DTL-83413, is not required to perform group II and group III First Article inspections on Grounding clamps. Group I inspections are still required. First article inspection shall consist of the inspections specified in table I.

Samples for first article inspection shall be representative of the products proposed to be furnished to this specification. Test samples shall consist of 100% testing, up to 12 grounding clamps. Test samples shall be examined and tested in accordance with MIL-DTL-83413 and this specification. Test results shall be submitted to the acquiring activity.



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TABLE I. First article inspection.

| Inspection | Plug | Receptacle | Clamp connector | Jumper assembly | Requirement paragraph | Test Paragraph |
|---|---|---|---|---|---|---|
| Group I | | | | | | |
| Visual and mechanical Inspection | | | X | | 3.1, 3.3, 3.4 3.6, 3.8 | 4.7.1 |
| Detent action | | | | | 3.5.1 | 4.7.2 |
| Handgrip force | | | X | | 3.5.2 | 4.7.3 |
| Side loading | | | | | 3.5.3 | 4.7.4 |
| Torque | | | | | 3.5.4 | 4.7.5 |
| Engagement | | | | | 3.5.5 | 4.7.6 |
| Contact resistance | | | | | 3.5.6.1 | 4.7.7.1 |
| Clamp connector resistance | | | X 2/ | | 3.5.6.2 | 4.7.7.2 |
| Assembly resistance | | | | | 3.5.6.3 | 4.7.7.3 |
| Group II | | | | | | |
| Cable attachment | | | X | | 3.5.7 | 4.7.8 |
| Clamp release force (initial) | | | X | | 3.5.8.1 | 4.7.9 |
| Clamp release force (after conditioning) | | | X | | 3.5.8.2 | 4.7.9 |
| Clamping life | | | X | | 3.5.9 | 4.7.10 |
| Vibration | | | | | 3.5.10 | 4.7.11 |
| Durability | | | | | 3.5.11 | 4.7.12 |
| Flexibility | | | | | 3.5.13 | 4.7.14 |
| Tensile strength | | | | | 3.5.12 | 4.7.13 |
| Crush | | | | | 3.5.14 | 4..7.15 |
| Group III | | | | | | |
| Humidity | | | X | | 3.5.15 | 4.7.16 |
| Temperature cycling | | | X | | 3.5.16 | 4.7.17 |
| Drop | | | X | | 3.5.17 | 4.7.18 |
| Altitude – low temperature | | | | | 3.5.18 | 4.7.19 |
| Dust | | | | | 3.5.19 | 4.7.20 |
| Fluid immersion | | | | | 3.5.20 | 4.7.21 |
| Salt spray | | | X | | 3.5.22 | 4.7.23 |
| Group IV | | | | | | |
| Lightning current | | | | | 3.5.23 | 4.7.24 |

1/ - X indicates that a test is applicable.
2/ - Clamp connector resistance after salt spray.



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Referenced documents. In addition to MIL-DTL-83413, this document references the following:

AIA/NASM51021
ASTM-A228/A228M
ASTM-B633
ASTM-B733
EIA/ECA-364-26
SAE-AMS 2700
SAE-AMS-5343
SAE-AMS-5355

CONCLUDING MATERIAL

Custodians:
Army - AV
Navy - AS
Air Force - 85
DLA - CC

Preparing activity:
DLA - CC

(Project 5999-2018-010)

Review activities:
Army - AR, CR, CR4, MI
Navy - MC
Air Force - 02

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.



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